



University of Central Punjab

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FACULTY OF INFORMATION TECHNOLOGY

2 – Input OR Gate Using Transistor

Project Description

Project Name	2 – Input OR Gate Using Transistor
Course Title	Basic Electronics - LAB
Submission Date	15-July-2025
Faculty	Faculty Of Information Technology
Software Used	Tinkercad

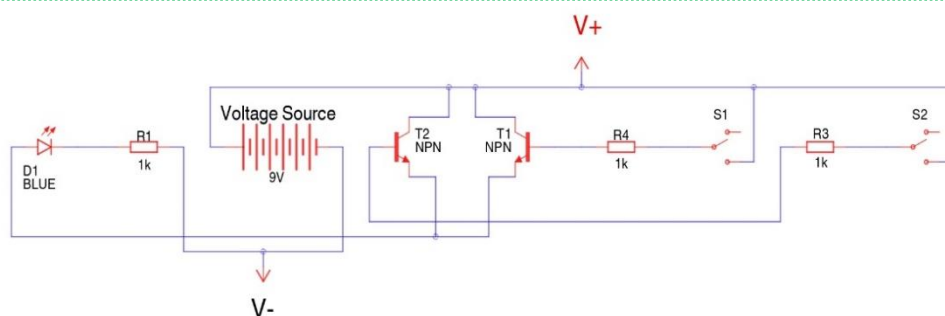


1. Executive Summary :

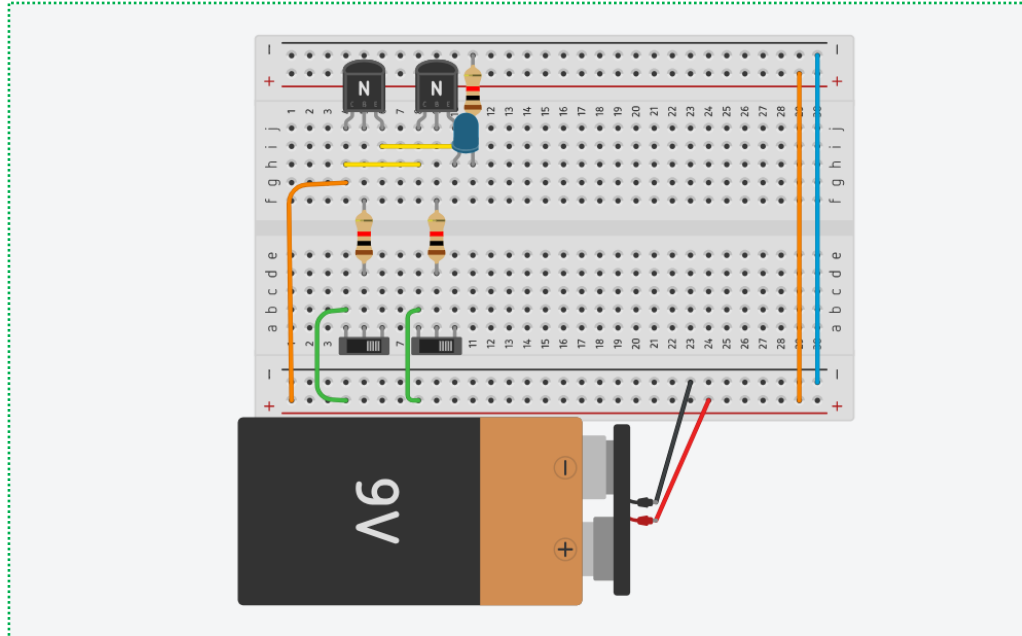
This project demonstrates the implementation of a **2-input OR logic gate** using **NPN transistors**. The primary objective was to design and test a basic OR gate using discrete electronic components rather than integrated circuits (ICs). This hands-on lab activity aimed to help students understand the behavior of logic gates at the component level, reinforcing concepts of digital electronics, transistor switching, and circuit logic behavior.

- The project successfully replicated the OR gate functionality: the output was **HIGH** (LED ON) when **either or both inputs were HIGH**, and **LOW** (LED OFF) when **both inputs were LOW**.

2. Methodology - Circuit Diagram :



Circuit Diagram On Tinkercad :



Components Used :

- **Breadboard**
- **2 NPN Transistors (T1 & T2)**
- **2 Switches (S1 & S2)** as inputs
- **Resistors (R1 to R3)** to limit base and LED current
- **1 Blue LED (D1)** for output visualization
- **9V DC** power source



3. Implementation Steps :

1. Connected **V+ (9V)** and **V- (GND)** as the main power supply.
2. Placed **two NPN transistors (T1 and T2)** in parallel at the output stage.
3. Connected the base of each transistor to individual **resistors (R3 & R4)**, followed by **input switches S1 and S2**.
4. Connected a **blue LED** in series with a **resistor (R1)** to the collector side of the transistors.
5. **Grounded** the emitters of both transistors.
6. **Tested the output** by sliding each input switch and observing LED behavior.

4. Results and Findings :

- If **no switch** is slide to on → **LED OFF**
- If **S1** is slide to on → **LED ON**
- If **S2** is slide to on → **LED ON**
- If **both** are slided to on → **LED ON**

This confirms the **OR gate behavior** using basic components.



Switch S1 (A)	Switch S2 (B)	LED Output (OR)
0 (OFF)	0 (OFF)	0 (OFF)
1 (ON)	0 (OFF)	1 (ON)
0 (OFF)	1 (ON)	1 (ON)
1 (ON)	1 (ON)	1 (ON)

5. Positive Outcomes :

- Built a fully working **2-input OR logic gate** using only basic components.
- Verified **OR logic truth table** using physical switches and observed output through an LED.
- Deepened understanding of **transistor switching and circuit-level** logic design.